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Local constraint based optimization for segmenting neuronal structures in 2D electron microscopy sections

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Abstract:
The method described here is for segmenting neuron slices from individual EM slices (2D).

Figure 1 shows an **EM image** obtained from serial section transmission electron microscopy (ssTEM) from the ventral nerve chord (VNC) of Drosophila larva.

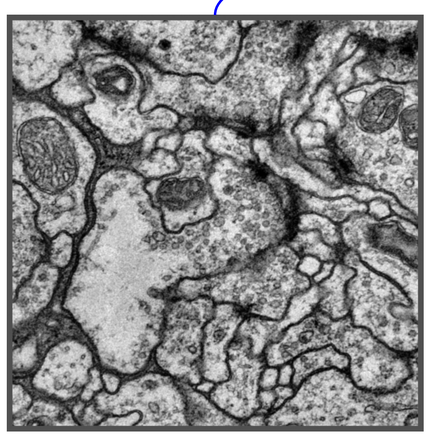

Figure 1: raw EM image

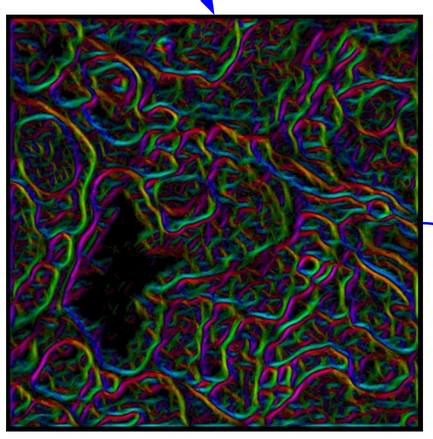

Figure 2: max response of oriented filters

Figure 2 is the result of applying **oriented edge detection filters** and plotting the maximum response at each point. The color codes for the orientation of the edge detector with the highest response.

Figure 4 is a graph that directly follows from figure 3. Each edge is a watershed ridge that separates two adjacent watershed regions. A node in the graph is where more than two such edges meet.

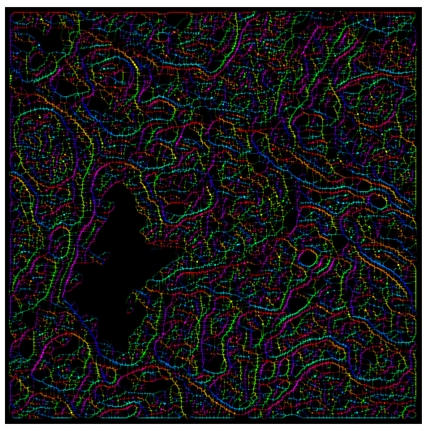

Figure 4: Graph obtained from the watershed transform

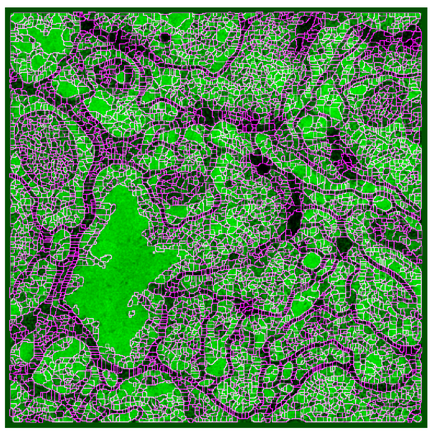

Figure 3: Watershed transform of figure 2.

Figure 3 is obtained by applying **Watershed Transformation** on figure 2. The pink edges are the watershed ridges.

Integer Linear Program (ILP)

ILP Variable types (*indicator variables*):
e – directional edge connecting 2 nodes
n – node
r – region (bounded by edges)

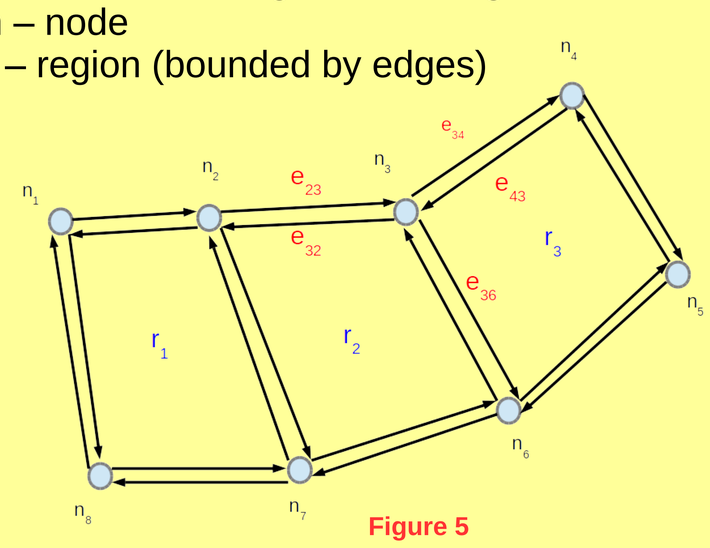

Figure 5

ILP constraints:

1. Edge activation	$\forall e_i \in \mathcal{E},\ e_{ij} + e_{ji} + e^0_{ij} = 1$
2. Node activation	$\forall n_i \in \mathcal{N},\ \sum_{j=0}^{k} n_i^{(j)} = 1$
3. Region activation	$\forall r_i \in \mathcal{R},\ r_i + r_i^{(0)} = 1$
4. Closed object boundaries	$\forall n_i \in \mathcal{N},\ 2n_i^{(0)} + \sum_{e_{ij} \in E_i} e_{ij} = 2$
5. Edge directionality at each node: one inwards, one outwards	$\forall n_i \in \mathcal{N},\ 0 \leq -2n_i^{(j)} + e_{ik} + e_{ki} \leq 1$
6. Edge direction and region state coupling	$\forall e_{ij} \in \mathcal{E},\ -e_{ij} + e_{ji} + r_j - r_j = 0$

ILP objective function:

$$\{e^*, n^*, r^*\} = \underset{e \in \mathcal{E}, n \in \mathcal{N}, r \in \mathcal{R}}{\operatorname{argmin}} \quad C_e + C_n + C_r$$

Figure 6 shows the *a priori* probability of each edge to be activated. These probabilities can be obtained from a trained classifier using annotated data. Brighter (white) pixel intensities show a probability closer to 1. These values are used to formulate the ILP objective function.

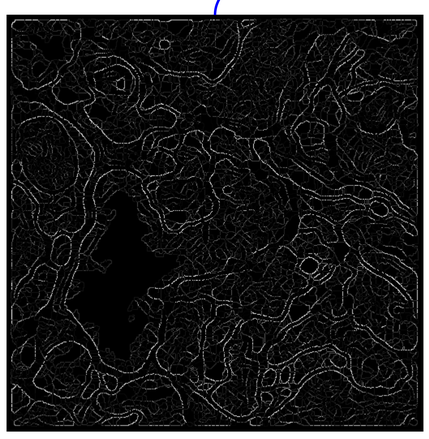

Figure 6: Edge unary score map

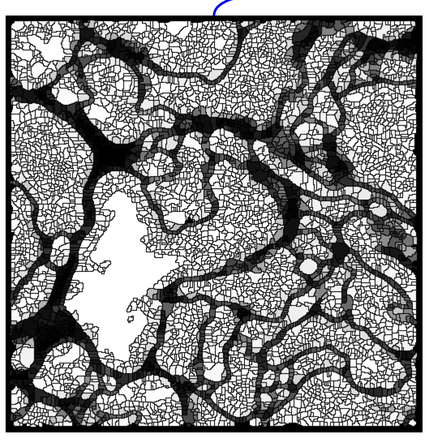

Figure 7: Region unary score map

Figure 7 shows the *a priori* probability of each watershed region to be foreground (neuron interior). These probabilities are obtained from a trained classifier using annotated data. Brighter (white) pixel intensities show a probability closer to 1. These values are used to formulate the ILP objective function.

Table 1 shows commonly used error measures for evaluating image segmentation methods, namely the *Rand index, Generalized Consistency Error (GCE)* and the *Variation of Information (VoI)* applied to the output of a test data set.

Rand Index	0.9408
GCE	0.1878
VoI	1.3840

Table 1: Performance measures

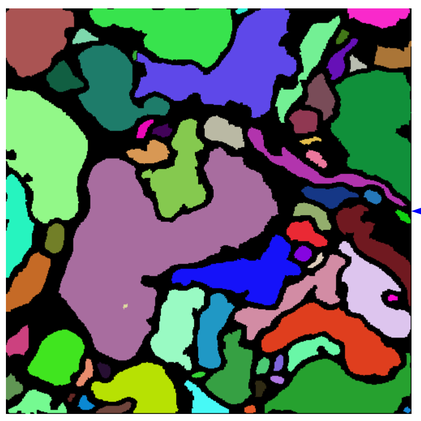

Figure 8: Neuron segmentation obtained by solving the ILP

Figure 8 shows the output segmentation our approach produced for the input image shown in Figure 1. Each neuron slice is visualized in a unique randomly assigned color.

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